PRODUCT DATA SHEET

DUROPROOF BENTONITE

Two Component HDPE / Bentonite
Waterproofing Membrane

DUROTECH
Engineered Construction Chemicals
Made In Australia For Over 45 Years

DESCRIPTION

Duroproof Bentonite is a multi-layer waterproofing membrane consisting of virgin high-density polyethylene, bentonite and a protective layer of nonwoven polypropylene. The tough HDPE membrane provides the primary waterproofing envelope while the expandable bentonite clay provides the self- sealing reassurance under hydrostatic conditions. It can expand up to 10 times its original thickness to prevent water that may make it past the HDPE layer, providing Bentonite protection. The polypropylene fabric protects the bentonite from exposure to inclement weather and direct installation of shotcrete.

USE AREAS

Duroproof Bentonite waterproofing membrane is used on structures below ground and is especially effective where the waterproofing is applied before the walls or floor are poured. Duroproof Bentonite is effective for blindside installation such as lagging, under floors and elevator pits or standard installations such as backfilled wall, decks/tunnels. It has outstanding performance when used under conditions of high water head.

Installation

For complete installation guidelines, please ask your Durotech Technical Representative or visit our website for details.

Preparatory Work

Duroproof Bentonite requires firm background to maintain 121kg per m2 compression. Fill voids or spaces more than 25mm with grout and/or plywood. Remove nails and sharp protrusions over 6.4mm. Installation may proceed on green concrete with damp or frozen surfaces in all weather but standing water must be removed. Cover lagging boards and soldier piles with suggested drainage layer. Complete installation at penetrations (utility and/or tiebacks) as described by the manufacturer.

TECHNICAL INFORMATION

Membrane		Green 20-mil virgin resin HDPE
Bentonite		Sodium Montmorillonite
Weight		4.89 kg/m²
Puncture Resistance	ASTM E154-88	170 lbs. (77.3 kg)
Tensile Strength: Membrane	ASTM D638	MD: 3660 psi (25.2 MPa) TD; 3650 psi (25.2 MPa)
% Elongation at break	ASTM D638 Type I dogbone	>700%
Resistance to hydrostatic head	ASTM 5385 ASTM D751 Procedure A	70m of water 52.9m of water
Vapour Water Permeability:	ASTM E96-80	0.53 x 10-13cm/sec 0.84 ng/ m2.s.Pa 0.033 Perms (grains/ft2* hr * inHg)
Resistance to micro-organisms: (bacteria, fungi, mold, yeast)	ASTM E154-88	Unaffected
Toxicity:		Low. Do not ingest
Staining:		No known incompatibilities
Chemical Resistance:		Extremely high resistance tochemicals & gases. Contactmanufacturer for specific information.
Freeze/thaw cycles:		No effect before or after installation.
Installation Temperatures	ASTM D746, ASTM D1238	-40˚C to 65.5˚C
Life Expectancy:		Both high-density polyethylene and bentonite have life expectancy measurable in thousands-of-years.

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Backfilled Wall

Duroproof Bentonite rolls are installed, either vertically or horizontally with the bentonite side towards the concrete structure, by nailing across the top every 50mm, lapping seams at least 75mm. Overlap seams as shingles. Close the seams with nails at 900mm centres and Seam Tape.

Lagging

Duroproof Bentonite rolls are installed, either vertically or horizontally with the bentonite side towards the concrete structure, by nailing across the top every 50mm, lapping seams at least 75mm. Overlap seams as shingles. Close the seams with nails at 600mm centres and box staple between.

Under Slab

Duroproof Bentonite waterproofing system will provide a waterproof seal and a vapor barrier. Roll out membrane bentonite-side down on the blinding slab or over compacted earth with 150micron HDPE. Overlap and stagger seams at least 75mm. Fasten with staples or nails. Tape seams. Tie in the floor to other surfaces as described by the manufacturer. Protect area from flooding prior to concrete pour.

Protection

Duroproof Bentonite waterproofing system does not require an additional protection course for most applications. For special applications, contact your Durotech Technical Representative for details.

Packaging

Protect from moisture. Store on skid or pallet, cover with polyethylene or tarp.

Storage

Roll Sizes: 1.1m x 6.6m or 7.26m2 standard rolls. Customised lengths are available by SPECIAL ORDER.

Limitations

Duroproof Bentonite installation must be confined by a minimum of 121kg per m2. Keep Duroproof Bentonite dry, protect from exposure to the elements. Duroproof Bentonite is resistant to many common contaminants in soil. Please contact Durotech for

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